Q11 Comprehension Check

Question: Kalai and Vempala argue that a *calibrated* language model trained on ideal (fully truthful) data must still hallucinate on "arbitrary" facts at a rate close to its monofact rate. A colleague concludes: "So hallucination is unavoidable — we should stop funding mitigation work." Identify the flaw(s) in the conclusion, citing (a) which facts the theorem covers, (b) what the theorem assumes about pre-training vs. post-training, and (c) what mitigation actually does to the reliability boundary.

Answer: The conclusion overreaches on all three points. (a) The lower bound applies to *arbitrary* facts — those whose truth cannot be determined from training data — and specifically *not* to systematic facts (arithmetic, rule-checkable claims) or frequently-attested facts like publication references, where the bound gives no force. (b) The theorem describes *pre-trained* models optimized to be good predictors; it explicitly does not preclude post-training that reduces hallucination — at the cost of degrading calibration, which is precisely the observed behavior of aligned models. (c) Mitigation does not need to move the statistical floor to be worthwhile: retrieval extends what counts as "attested" (converting arbitrary facts into context-supported ones), abstention and post-training shift behavior toward the reliability boundary (fewer samples beyond it), and verification filters samples after the fact. The theorem says some floor exists for a specific fact class under a specific training regime — not that engineering cannot save the workloads we actually care about.